

ROAS and VLAN lab exercise

In this lab exercise you will have to configure a Router-On-a-Stick (ROAS) and a switch with three different VLANs. All basic configuration of the devices has been completed and does not have to be done.

1. On SW1:
 - a. Configure a VLAN named “HR” with the number 10.
 - b. Configure a VLAN named “Administrative” with the number 20.
 - c. Configure a VLAN named “Tehnnical” with the number 30.
 - d. Configure F0/1 as an access port with VLAN 10.
 - e. Configure F0/2 as an access port with VLAN 20.
 - f. Configure F0/3 as an access port with VLAN 30.
 - g. Configure G0/1 as a trunk port. Encapsulation type is not needed to be configured as only Dot1Q is allowed on the switch.
2. On R1:
 - a. Configure 3 subinterfaces of interface G0/1.
 - b. Assign the matching Dot1Q VLAN ID to each subinterface.
 - c. Use each of the PC's default gateway as the subinterface IP address.
Use subnet mask /24.
3. Ping PC3 from PC1. The first ping is most likely going to fail – the rest should respond.
4. On SW1 trunk interface, allow only VLANs 10 and 20. Try to ping PC3 from PC1 – this shouldn't work.
5. Ping PC1 from PC2.